



MD880-61 Series Single-phase Braking Unit

Parameter Guide



Industrial
Automation



Intelligent
Elevator



New Energy
Vehicle



Industrial
Robot



Rail
Transit



Data code 19012489A00

Preface

Introduction

The MD880-61M series three-phase braking module provides comprehensive protective functions, including protections against overcurrent, overvoltage, overtemperature, braking unit open circuit, and short circuit. It allows you to set parameters including braking start/stop voltage and braking rate. Multiple braking modules can be deployed in master/slave mode to meet the braking requirements of high power starters. When the drive brakes, the kinetic energy is converted to the electric energy which is fed back to the drive due to the mechanical inertia of the load. This will cause rising of the DC bus voltage of the drive. To ensure normal operation of the drive, a braking unit is required. Therefore, the regenerative energy can be consumed by the braking resistor, preventing the drive from stop due to overvoltage protection. The MD880 series single-phase braking units mainly apply to applications where AC drives or servo drives (hereinafter referred to as drives) require rapid speed reduction, positioning, and braking.

More Data

Name	Data Code	Description
MD880-61 Series Single-phase Braking Unit User Guide	19012239	Presents the product information, mechanical installation, electrical installation, and routine maintenance.

Revision History

Date	Version	Software Version	Revision
July 4 2024	A00	Version: A04 Version label: A2-10 = 1.02	First release

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1 Parameter List

A0: Status monitoring

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-00	Current state machine	16603	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Invalid 1: Initializing 2: Fault 3: Switch-on blocked 4: Ready for switch-on 5: Ready to run 6: Forward run 7: Test run							
	Description: Displays the current operating state of the brake.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-02	Braking current active value	16605	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: Displays the active value of the braking current.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-03	DC bus voltage	16606	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: Displays the DC bus voltage value.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

A0-04	Braking current peak value	16607	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: Displays the peak value of braking current.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-05	Braking power	16608	0.0 kW to 6553.5 kW	0.0 kW	UInt16	N/A	Standard user	-
	Description: Displays current braking power.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-08	Braking rate	16611	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: Displays current braking rate.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-11	IGBT module 1 temperature	16614	-32767°C to +32767°C	0°C	Int16	N/A	Standard user	-
	Description: Displays current temperature of braking transistor 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A0-12	IGBT module 2 temperature	16615	-32767°C to +32767°C	0°C	Int16	N/A	Standard user	-
	Description: Displays current temperature of braking transistor 2.							

A1: Faults and Alarms

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-00	Master code of current fault 1	16458	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-01	Subcode of current fault 1	16459	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-02	Information of current fault 1	16460	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-03	Master code of current fault 2	16461	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current fault 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-04	Subcode of current fault	16462	0 to 65535	0	UInt16	N/A	Standard user	-

	2							
	Description: Displays the subcode of current fault 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-05	Information of current fault 2	16463	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current fault 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-06	Master code of current fault 3	16464	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current fault 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-07	Subcode of current fault 3	16465	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current fault 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-08	Information of current fault 3	16466	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current fault 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-09	Master code of current fault 4	16467	0 to 65535	0	UInt16	N/A	Standard user	-

	Description: Displays the master code of current fault 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-10	Subcode of current fault 4	16468	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current fault 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-11	Information of current fault 4	16469	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current fault 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-12	Master code of current fault 5	16470	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current fault 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-13	Subcode of current fault 5	16471	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current fault 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-14	Information of current fault 5	16472	0 to 65535	0	UInt16	N/A	Standard user	-
	Description:							

	Displays the information of current fault 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-15	Master code of current fault 6	16473	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current fault 6.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-16	Subcode of active fault code 6	16474	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current fault 6.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-17	Information of current fault 6	16475	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current fault 6.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-36	Master code of current alarm code 1	16476	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current alarm 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-37	Subcode of current alarm 1	16477	0 to 65535	0	UInt16	N/A	Standard user	-
	Description:							

	Displays the subcode of current alarm 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-38	Information of current alarm 1	16478	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current alarm 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-39	Master code of current alarm code 2	16479	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current alarm 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-40	Subcode of current alarm 2	16480	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current alarm 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-41	Information of current alarm 2	16481	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current alarm 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-42	Master code of current alarm code 3	16482	0 to 65535	0	UInt16	N/A	Standard user	-

	Description: Displays the master code of current alarm 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-43	Subcode of current alarm 3	16483	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current alarm 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-44	Information of current alarm 3	16484	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current alarm 3.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-45	Master code of current alarm code 4	16485	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current alarm 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-46	Subcode of current alarm 4	16486	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current alarm 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-47	Information of current alarm 4	16487	0 to 65535	0	UInt16	N/A	Standard user	-

	Description: Displays the information of current alarm 4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-48	Master code of current alarm 5	16488	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current alarm 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-49	Subcode of current alarm 5	16489	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the subcode of current alarm 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-50	Information of current alarm 5	16490	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current alarm 5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-51	Master code of current alarm 6	16491	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the master code of current alarm 6.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-52	Subcode of current alarm 6	16492	0 to 65535	0	UInt16	N/A	Standard user	-
	Description:							

	Displays the subcode of current alarm 6.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A1-53	Information of current alarm 6	16493	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the information of current alarm 6.							

A2: Drive information

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-00	Product type	16512	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 7: Single-phase braking							
	Description: Displays the current device type.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-01	Module rated power	16513	0.0 kW to 6553.5 kW	0.0 kW	UInt16	N/A	Standard user	-
	Description: Displays the rated power of the braking unit.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-02	Module rated voltage	16514	0 V to 65535 V	0 V	UInt16	N/A	Standard user	-
	Description: Displays the rated voltage of the braking unit.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

A2-03	Module rated current	16515	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: Displays the rated current of the braking unit.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-09	Software type	16536	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Standard software 1: Customized software							
	Description: Displays the current software type.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-10	Main software version	16537	0.00 to 655.35	0.00	UInt16	N/A	Standard user	-
	Description: Displays the main version of current software.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A2-23	Parameter version	16545	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Displays the version of current parameters.							

A8: Environment setting

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A8-02	Parameter access level	65	See "Value".	1	UInt16	Real time	Standard user	-
	Value:							

	0: Monitoring parameter 1: Basic parameter 2: Expert parameter 3: Maintenance parameter							
	Description: Used to set the level of the parameter currently displayed. To set a higher parameter level, enter the manufacturer password through A8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A8-03	Manufacturer password	66	0 to 65535	0	UInt16	Real time	Standard user	-
	Description: Used to set the default password. Contact the technical support.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
A8-08	Parameters copied from RAM to ROM	68	See "Value".	0	UInt16	At stop	Standard user	-
	Value: 0: No action 1: Save all parameters							
	Description: The parameters are saved in FLASH in the main control chip, which are retentive upon power failure.							

H2: Current fault log

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-00	Master code of current fault 1	269	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Records the master code of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

H2-01	Subcode of current fault 1	270	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Records the subcode of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-02	Information of current fault 1	271	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: Records the information of current fault 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-03	Master code of current fault 2	272	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-00.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-04	Subcode of current fault 2	273	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-01.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-05	Information of current fault 2	274	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-02.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-06	Master code	275	0 to 65535	0	UInt16	N/A	Standard	-

	of current fault 3						user	
	Description: See the description of H2-00.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-07	Subcode of current fault 3	276	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-01.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-08	Information of current fault 3	277	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-02.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-09	Master code of current fault 4	278	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-00.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-10	Subcode of current fault 4	279	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-01.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-11	Information of current	280	0 to 65535	0	UInt16	N/A	Standard user	-

	fault 4							
	Description: See the description of H2-02.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-12	Master code of current fault 5	281	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-00.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-13	Subcode of current fault 5	282	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-01.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-14	Information of current fault 5	283	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-02.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-15	Master code of current fault 6	284	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-00.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-16	Subcode of current fault 6	285	0 to 65535	0	UInt16	N/A	Standard user	-

	Description: See the description of H2-01.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-17	Information of current fault 6	286	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the description of H2-02.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-18	DC bus voltage upon current fault	287	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: Records the DC bus voltage upon occurrence of current fault.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-19	Braking current upon current fault	288	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: Records the braking current upon occurrence of current fault							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-20	Braking rate upon current fault	289	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: Records the braking rate upon occurrence of current fault							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-21	Module 1 temperature	290	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-

	upon current fault							
	Description: Records the temperature of module 1 upon occurrence of current fault.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-22	Module 2 temperature upon current fault	291	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: Records the temperature of module 2 upon occurrence of current fault.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-23	Status word upon current fault	292	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: Records the system status word upon occurrence of current fault.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-24	Accumulative power-on hours upon current fault	293	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-
	Description: Records the system operation time in hours upon occurrence of current fault.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-25	Accumulative power-on seconds upon current fault	294	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: Records the system operation time in seconds upon occurrence of current fault.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H2-26	Reserved	295	0 to 65535	0	UInt16	N/A	Standard user	-
H2-27	Starting voltage for braking upon current fault	296	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: Records the starting voltage for braking upon occurrence of current fault.							

H3: Last fault log

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-00	Master code of the last fault 1	301	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-01	Subcode of the last fault 1	302	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-02	Information of the last fault 1	303	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-03	Master code of the last fault 2	304	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-04	Subcode of the last fault 2	305	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-05	Information of the last fault 2	306	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-06	Master code of the last fault 3	307	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-07	Subcode of the last fault 3	308	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
H3-08	Information of the last fault 3	309	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-09	Master code of the last fault 4	310	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-10	Subcode of the last fault 4	311	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-11	Information of the last fault 4	312	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-12	Master code of the last fault 5	313	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

H3-13	Subcode of the last fault 5	314	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-14	Information of the last fault 5	315	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-15	Master code of the last fault 6	316	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-16	Subcode of the last fault 6	317	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-17	Information of the last fault 6	318	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-18	DC bus	319	0.0 V to	0.0 V	UInt16	N/A	Standard	-

	voltage upon the last fault		6553.5 V				user	
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-19	Braking current upon the last fault	320	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-20	Braking rate upon the last fault	321	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-21	Module 1 temperature upon the last fault	322	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-22	Module 2 temperature upon the last fault	323	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-23	Status upon the last fault	324	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-24	Accumulative power-on hours upon the last fault	325	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-25	Accumulative power-on seconds upon the last fault	326	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H3-26	Reserved	327	0 to 65535	0	UInt16	N/A	Standard user	-
H3-27	Starting voltage for braking upon last fault	328	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

H4: 2nd to last fault log

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-00	Master code upon the 2nd to last fault 1	333	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-01	Subcode of the 2nd to last fault 1	334	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-02	Information of the 2nd to last fault 1	335	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-03	Master code of the 2nd to last fault 2	336	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-04	Subcode of the 2nd to	337	0 to 65535	0	UInt16	N/A	Standard	-

	last fault 2						user	
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-05	Information of the 2nd to last fault 2	338	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-06	Master code of the 2nd latest fault 3	339	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-07	Subcode of the 2nd to last fault 3	340	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-08	Information of the 2nd to last fault 3	341	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-09	Master code of the 2nd to last fault 4	342	0 to 65535	0	UInt16	N/A	Standard user	-

	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-10	Subcode of the 2nd to last fault 4	343	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-11	Information of the 2nd to last fault 4	344	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-12	Master code of 2nd to last fault 5	345	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-13	Subcode of the 2nd to last fault 5	346	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-14	Information of the 2nd to last fault 5	347	0 to 65535	0	UInt16	N/A	Standard user	-
	Description:							

	See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-15	Master code of the 2nd to last fault 6	348	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-16	Subcode of the 2nd to last fault 6	349	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-17	Information of the 2nd to last fault 6	350	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-18	DC bus voltage upon the 2nd to last fault	351	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-19	Braking current upon the	352	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-

	2nd to last fault							
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-20	Braking rate upon the 2nd to last fault	353	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-21	Module 1 temperature upon the 2nd to last fault	354	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-22	Module 2 temperature upon the 2nd to last fault	355	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-23	Status word of the 2nd to last fault	356	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-24	Accumulative power-on hours upon the 2nd to last fault	357	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-25	Accumulative power-on seconds upon the 2nd to last fault	358	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H4-26	Reserved	359	0 to 65535	0	UInt16	N/A	Standard user	-
H4-27	Starting voltage for braking upon the 2nd to last fault	360	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

H5: 3rd to last fault log

Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective
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		tion address	range			mode	authority	time
H5-00	Master code of the 3rd to last fault 1	365	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-01	Subcode of the 3rd to last fault 1	366	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-02	Information of the 3rd to last fault 1	367	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-03	Master code of the 3rd to last fault 2	368	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-04	Subcode of the 3rd to last fault 2	369	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time

H5-05	Information of the 3rd to last fault 2	370	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-06	Master code of the 3rd to last fault 3	371	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-07	Subcode of the 3rd to last fault 3	372	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-08	Information of the 3rd to last fault 3	373	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-09	Master code of the 3rd to last fault 4	374	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-10	Subcode of	375	0 to 65535	0	UInt16	N/A	Standard	-

	the 3rd to last fault 4						user	
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-11	Information of the 3rd to last fault 4	376	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-12	Master code of the 3rd to last fault 5	377	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-13	Subcode of the 3rd to last fault 5	378	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-14	Information of the 3rd to last fault 5	379	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-15	Master code of the 3rd to	380	0 to 65535	0	UInt16	N/A	Standard user	-

	last fault 6							
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-16	Subcode of the 3rd to last fault 6	381	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-17	Information of the 3rd to last fault 6	382	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-18	DC bus voltage upon the 3rd to last fault	383	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-19	Braking current upon the 3rd to last fault	384	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
H5-20	Braking rate upon the 3rd to last fault	385	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica- tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-21	Module 1 temperature upon the 3rd to last fault	386	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica- tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-22	Module 2 temperature upon the 3rd to last fault	387	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica- tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-23	Status word of the 3rd to last fault	388	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica- tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-24	Accumulative power-on hours upon	389	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-

	the 3rd to last fault							
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-25	Accumulative power-on seconds upon the 3rd to last fault	390	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H5-26	Reserved	391	0 to 65535	0	UInt16	N/A	Standard user	-
H5-27	Starting voltage for braking upon the 3rd to last fault	392	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

H6: 4th to last fault log

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-00	Master code of the 4th to last fault 1	397	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-01	Subcode of the 4th to last fault 1	398	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-02	Information of the 4th to last fault 1	399	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-03	Master code of the 4th to last fault 2	400	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-04	Subcode of the 4th to last fault 2	401	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-05	Information of the 4th to last fault 2	402	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
H6-06	Master code of the 4th to last fault 3	403	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-07	Subcode of the 4th to last fault 3	404	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-08	Information of the 4th to last fault 3	405	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-09	Master code of the 4th to last fault 4	406	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-10	Subcode of the 4th to last fault 4	407	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time

H6-11	Information of the 4th to last fault 4	408	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-12	Master code of the 4th to last fault 5	409	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-13	Subcode of the 4th to last fault 5	410	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-14	Information of the 4th to last fault 5	411	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-15	Master code of the 4th to last fault 6	412	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-16	Subcode of	413	0 to 65535	0	UInt16	N/A	Standard	-

	the 4th to last fault 6						user	
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-17	Information of the 4th to last fault 6	414	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-18	DC bus voltage upon the 4th to last fault	415	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-19	Braking current upon the 4th to last fault	416	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-20	Braking rate upon the 4th to last fault	417	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-21	Module 1 temperature upon the 4th to last fault	418	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-22	Module 2 temperature upon the 4th to last fault	419	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-23	Status word of the 4th to last fault	420	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-24	Accumulative power-on hours upon the 4th to last fault	421	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

H6-25	Accumulative power-on seconds upon the 4th to last fault	422	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H6-26	Reserved	423	0 to 65535	0	UInt16	N/A	Standard user	-
H6-27	Starting voltage for braking upon the last to 4th fault	424	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

H7: 5th to last fault log

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-00	Master code of the 5th to last fault 1	429	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-01	Subcode of the 5th to last fault 1	430	0 to 65535	0	UInt16	N/A	Standard user	-
	Description:							

	See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-02	Information of the 5th to last fault 1	431	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-03	Master code of the 5th to last fault 2	432	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-04	Subcode of the 5th to last fault 2	433	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-05	Information of the 5th to last fault 2	434	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-06	Master code of the 5th to last fault 3	435	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-07	Subcode of the 5th to last fault 3	436	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-08	Information of the 5th to last fault 3	437	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-09	Master code of the 5th to last fault 4	438	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-10	Subcode of the 5th to last fault 4	439	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-11	Information of the 5th to last fault 4	440	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
H7-12	Master code of the 5th to last fault 5	441	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-13	Subcode of the 5th to last fault 5	442	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-14	Information of the 5th to last fault 5	443	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-15	Master code of the 5th to last fault 6	444	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-16	Subcode of the 5th to last fault 6	445	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time

H7-17	Information of the 5th to last fault 6	446	0 to 65535	0	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-18	DC bus voltage upon the 5th to last fault	447	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-19	Braking current upon the 5th to last fault	448	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-20	Braking rate upon the 5th to last fault	449	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-21	Module 1 temperature upon the 5th to last	450	-32768°C to +3276°C	0°C	Int16	N/A	Standard user	-

	fault							
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-22	Module 2 temperature upon the 5th to last fault	451	-32768°C to +32676°C	0°C	Int16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-23	Status word of the 5th to last fault	452	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-24	Accumulative power-on hours upon the 5th to last fault	453	0 h and 65535 h	0 h	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-25	Accumulative power-on seconds upon the 5th to last fault	454	0 to 65535	0s	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

	See the descriptions for H2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H7-26	Reserved	455	0 to 65535	0	UInt16	N/A	Standard user	-
H7-27	Starting voltage for braking upon the 5th to last fault	456	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
	Description: See the descriptions for H2.							

H8 Monitoring Parameters

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-03	Data 1 braking rate	16687	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: Used to monitor the braking rate of data 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-04	Data 1 status word	16703	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: Used to monitor the monitoring status word of data 1.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-07	Data 2 braking rate	16688	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-08	Data 2 status word	16704	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-11	Data 3 braking rate	16689	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-12	Data 3 status word	16705	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-15	Data 4 braking rate	16690	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-16	Data 4 status word	16706	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-19	Data 5 braking rate	16691	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description:							

	See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-20	Data 5 status word	16707	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-23	Data 6 braking rate	16692	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-24	Data 6 status word	16708	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-27	Data 7 braking rate	16693	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-28	Data 7 status word	16709	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-31	Data 8 braking rate	16694	0 to 6553.5	0%	UInt16	N/A	Standard user	-

	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-32	Data 8 status word	16710	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-35	Data 9 braking rate	16695	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-36	Data 9 status word	16711	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-39	Data 10 braking rate	16696	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-40	Data 10 status word	16712	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-43	Data 11	16697	0 to 6553.5	0%	UInt16	N/A	Standard	-

	braking rate						user	
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-44	Data 11 status word	16713	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-47	Data 12 braking rate	16698	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-48	Data 12 status word	16714	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-51	Data 13 braking rate	16699	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-52	Data 13 status word	16715	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

H8-55	Data 14 braking rate	16700	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-56	Data 14 status word	16716	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-59	Data 15 braking rate	16701	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-60	Data 15 status word	16717	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-63	Data 16 braking rate	16702	0 to 6553.5	0%	UInt16	N/A	Standard user	-
	Description: See the description of H8-03.							
Param.	Name	Communica tion address	Setting range	Default	Data type	Change mode	User authority	Effective time
H8-64	Data 16 status word	16718	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
	Description: See the description of H8-04.							

U0: Bit connector 1

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U0-00	Logic 0	30720	0 to 1	0	Boolean	N/A	Standard user	-
U0-01	Logic 1	30721	0 to 1	0	Boolean	N/A	Standard user	-
U0-02	DI	30722	0 to 1	0	Boolean	N/A	Standard user	-
U0-03	RO	30723	0 to 1	0	Boolean	N/A	Standard user	-
U0-06	DI inversion	30726	0 to 1	0	Boolean	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U0-07	RO invert	30727	0 to 1	0	Boolean	N/A	Standard user	-
U0-10	Original DI state	30730	0 to 1	0	Boolean	N/A	Standard user	-
U0-11	Original RO state	30731	0 to 1	0	Boolean	N/A	Standard user	-

U1: Bit connector 2

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-00	Ready to switch on	30820	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 系统状态机是否大于等于 S4。							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time

U1-01	Switch-on blocked	30821	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Check whether system state machine is lower than S4.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-02	Ready to run	30822	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Check whether system state machine is higher than or equal to S5.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-03	Forward run	30823	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Indicates the operating status of the power transistor.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-04	Not running	30824	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Used to invert the state of U1-03.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-05	Fault activated	30825	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 0: No fault 1: Fault occurred							
	Description: Indicates whether a fault occurs.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-06	Alarm activated	30826	See "Value".	0	Boolean	N/A	Standard user	-
	Value:							

	0: No alarm 1: Alarm occurred							
	Description: Indicates whether an alarm occurs.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-07	No fault	30827	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 0: Fault occurred 1: No fault							
	Description: Used to invert the state of U1-05.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-08	No alarm	30828	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 1: Alarm occurred 1: No alarm							
	Description: Used to invert the state of U1-06.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-09	DC bus charged flag	30829	0 to 1	0	Boolean	N/A	Standard user	-
	Description: The DC bus voltage exceeds the undervoltage threshold. A4-12 indicates the undervoltage threshold of command channel 1. A6-08 indicates the undervoltage threshold of command channel 2.							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-10	Zero drift detection on braking current completed	30830	0 to 1	0	Boolean	N/A	Standard user	-

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-11	OFF2 activated	30831	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 0: Activated 1: Not activated							
	Description: Disable output							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-12	Braking enable	30832	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 0: Disable 1: Enable							
	Description: Used to set whether the DI is "active low".							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-13	Braking resistor detection completion flag	30833	0 to 1	0	Boolean	N/A	Standard user	-
U1-14	Fan control command	30834	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Used to set the fan control state. The setpoint 1 indicates "Running".							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-15	FLASH operation busy	30835	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Indicates whether FLASH is in the process of writing or erasing.							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-16	Current channel of braking standard mode	30836	See "Value".	0	Boolean	N/A	Standard user	-
	Value: 0: Channel 1 1: Channel 2							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-17	RS485 communication timeout	30837	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 82-1 RS485 communication timeout							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-18	Reserved	30838	0 to 1	0	Boolean	N/A	Standard user	-
U1-19	Reserved	30839	0 to 1	0	Boolean	N/A	Standard user	-
U1-20	Current sampling zero drift too large	30840	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 1-1 Current detection circuit error flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-21	Overvoltage	30841	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 5-1 Excessively high DC bus voltage flag							
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
U1-22	Hardware overcurrent	30842	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 2-1 Hardware overcurrent flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-23	Software overcurrent	30843	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 2-2 Software overcurrent flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-24	Drive overload	30844	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 10-1 AC drive overload flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-25	Braking resistor overload	30845	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 11-1 Braking resistor overload flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-26	Braking resistor deviation too large	30846	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 22-1 Excessively high braking resistance and 22-2 Excessively low braking resistance flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-27	Resistance of braking	30847	0 to 1	0	Boolean	N/A	Standard user	-

	resistor too high							
	Description: 22-1 Excessively high braking resistor flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-28	Resistance of braking resistor too low	30848	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 22-2 Excessively low braking resistor flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-29	IGBT module 1 overtemperature	30849	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 14-1 Module 1 overtemperature flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-30	IGBT module 2 overtemperature	30850	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 14-2 Module 2 overtemperature flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-31	IGBT module 1 temperature too low	30851	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 59-1 Excessively low module 1 temperature flag							

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-32	IGBT module 2 temperature too low	30852	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 59-2 Excessively low module 2 temperature flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-33	External Fault	30853	0 to 1	0	Boolean	N/A	Standard user	-
	Description: External fault flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-34	External alarm	30854	0 to 1	0	Boolean	N/A	Standard user	-
	Description: External alarm flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-35	Customized fault	30855	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Customized fault flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-36	Customized alarm	30856	0 to 1	0	Boolean	N/A	Standard user	-
	Description: Customized alarm flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-37	Continuous braking	30857	0 to 1	0	Boolean	N/A	Standard user	-

	timeout							
	Description: 117-1 Continuous braking of the brake timeout flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-38	Braking not enabled (DI)	30858	0 to 1	0	Boolean	N/A	Standard user	-
	Description: 15-3 Brake enable (DI) not activated flag							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U1-39	Reserved	30859	0 to 1	0	Boolean	N/A	Standard user	-
U1-40	Reserved	30860	0 to 1	0	Boolean	N/A	Standard user	-
U1-41	DIP switch status of the master/slave	30861	0 to 1	0	Boolean	N/A	Standard user	-
U1-42	DIP switch status of the master/slave (filter)	30862	0 to 1	0	Boolean	N/A	Standard user	-

U5: Word connector 1

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-00	DC bus voltage (filter)	24576	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
U5-01	Braking current peak value	24577	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
U5-02	Braking	24578	0.0 kW to	0.0 kW	UInt16	N/A	Standard	-

	power		6553.5 kW				user	
U5-03	DC bus voltage	24579	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
U5-04	Braking rate	24580	0 to 6553.5	0%	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-05	Working mode	24581	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Standard braking 1: Master/Slave mode 2: Auto-detection mode 3: Test mode							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-06	Reserved	24582	0 to 65535	0	UInt16	N/A	Standard user	-
U5-07	Reserved	24583	0 to 65535	0	UInt16	N/A	Standard user	-
U5-08	IGBT module 1 temperature	24584	-32767°C to +32767°C	0°C	Int16	N/A	Standard user	-
U5-09	IGBT module 2 temperature	24585	-32767°C to +32767°C	0°C	Int16	N/A	Standard user	-
U5-15	Master code of current fault	24591	0 to 65535	0	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-16	Subcode of current fault	24592	0 to 65535	0	UInt16	N/A	Standard user	-
U5-19	Master code of current alarm	24595	0 to 65535	0	UInt16	N/A	Standard user	-
U5-20	Subcode of	24596	0 to 65535	0	UInt16	N/A	Standard	-

	current alarm						user	
U5-21	Internal status word 1	24597	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-22	Internal status word 2	24598	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-23	Internal status word 3	24599	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-24	Internal status word 4	24600	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-25	SOP-20 command word	24601	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-26	IDS command word	24602	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-27	System status word	24603	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-28	Monitoring status word	24604	0x0000 to 0xFFFF	0x0000	UInt16	N/A	Standard user	-
U5-29	Reserved	24605	0 to 65535	0	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-30	Current state machine	24606	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Invalid 4: Ready for switch-on							

	1: Initializing 2: Fault 3: Switch-on blocked				5: Ready to run 6: Forward run 7: Test run			
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-31	Last state machine	24607	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Invalid 1: Initializing 2: Fault 3: Switch-on blocked 4: Ready for switch-on 5: Ready to run 6: Forward run 7: Test run							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-32	2nd to last state machine	24608	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Invalid 1: Initializing 2: Fault 3: Switch-on blocked 4: Ready for switch-on 5: Ready to run 6: Forward run 7: Test run							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-33	3rd to last state machine	24609	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: Invalid 1: Initializing 2: Fault 3: Switch-on blocked 4: Ready for switch-on 5: Ready to run 6: Forward run 7: Test run							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-34	4th to last state machine	24610	See "Value".	0	UInt16	N/A	Standard user	-

	Value: 0: Invalid 1: Initializing 2: Fault 3: Switch-on blocked 4: Ready for switch-on 5: Ready to run 6: Forward run 7: Test run							
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-35	Zero drift of braking current	24611	0.000 A to 65.535 A	0.000 A	UInt16	N/A	Standard user	-
U5-36	AD sampling voltage of DC bus voltage	24612	0.000 V to 65.535 V	0.000 V	UInt16	N/A	Standard user	-
U5-37	AD sampling voltage of braking current	24613	0.000 V to 65.535 V	0.000 V	UInt16	N/A	Standard user	-
U5-38	AD sampling voltage of DC bus voltage	24614	0 to 65535	0	UInt16	N/A	Standard user	-
U5-39	AD sampling value of braking current	24615	0 to 65535	0	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-40	AD sampling value of module temperature 1	24616	0 to 65535	0	UInt16	N/A	Standard user	-
U5-41	AD sampling value of module temperature 2	24617	0 to 65535	0	UInt16	N/A	Standard user	-

U5-42	AD sampling value based on 0 V	24618	0 to 65535	0	UInt16	N/A	Standard user	-
U5-43	AD sampling value based on 1.5 V	24619	0 to 65535	0	UInt16	N/A	Standard user	-
U5-44	Original value of braking current	24620	0.0 A to 6553.5 A	0.0 A	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U5-45	Temperature NTC resistance of IGBT module 1	24621	0 Ω to 65535 Ω	0 Ω	UInt16	N/A	Standard user	-
U5-46	Temperature NTC resistance of IGBT module 2	24622	0 Ω to 65535 Ω	0 Ω	UInt16	N/A	Standard user	-
U5-47	Voltage value after correction of IGBT module 1 temperature	24623	0.000 V to 65.535 V	0.000 V	UInt16	N/A	Standard user	-
U5-48	Voltage value after correction of IGBT module 2 temperature	24624	0.000 V to 65.535 V	0.000 V	UInt16	N/A	Standard user	-
U5-49	DC bus voltage (filtered)	24625	0.0 V to 6553.5 V	0.0 V	UInt16	N/A	Standard user	-
Param.	Name	Communication address	Setting	Default	Data type	Change	User	Effective

		tion address	range			mode	authority	time
U5-50	Parameter setting error code	24626	See "Value".	0	UInt16	N/A	Standard user	-
	Value: 0: The parameter setting is proper. 1: The current at each point of the braking resistor overload curve is not incremental. 2: The maximum braking rate of the braking curve of channel 2 (A6.06) is lower than the initial braking rate (A6.02). 3: The maximum braking rate voltage threshold of the braking curve of channel 2 (A6.04/A6.05) is lower than the starting voltage (A6.01).							

U9: Double-word connector 1

Param.	Name	Communication address	Setting range	Default	Data type	Change mode	User authority	Effective time
U9-00	Duration of operation	28672	0s to 4294901760s	0s	UInt32	N/A	Standard user	-
U9-01	Accumulative operation time	28674	0s to 4294901760s	0s	UInt32	N/A	Standard user	-
U9-02	Power-on time of main circuit of current operation	28676	0s to 4294901760s	0s	UInt32	N/A	Standard user	-
U9-03	Accumulative power-on time of main circuit	28678	0s to 4294901760s	0s	UInt32	N/A	Standard user	-

2. List of Fault codes

Fault code	Fault name	Fault level	Default fault level	Reset method
1-1	Current detection circuit error	0: Blocked	0: Blocked	Reset manually
2-1	Hardware overcurrent	0: Blocked	0: Blocked	Reset by power-off
2 to 2	Software overcurrent	0: Blocked	0: Blocked	Reset by power-off/Reset manually
5-1	DC bus voltage too high	0: Blocked	0: Blocked	Reset automatically
10-1	Drive overload	0: Blocked	0: Blocked	Reset automatically
11-1	Braking resistor overload	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
14-1	Module 1 overtemperature	0: Blocked	0: Blocked	Reset automatically
14 to 2	Module 2 overtemperature	0: Blocked	0: Blocked	Reset automatically
15-1	External fault 1	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
15 to 2	External fault 2	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
15 to 3	Brake enable (DI) not activated	0: Blocked	0: Blocked	Reset automatically/Reset by power-off
22-1	Braking resistance too high	0: Blocked	0: Blocked	Reset by power-off
22-2	Braking resistance too low	0: Blocked	0: Blocked	Reset by power-off
22-3	Braking resistor detection loss	0: Blocked	0: Blocked	Reset by power-off/Reset automatically
27-1	Customized fault 1	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
27 to 2	Customized fault 2	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
27 to 3	Customized fault 3	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
27-4	Customized fault 4	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
28-1	Customized alarm 1	0: Blocked 5: Display alarm only	5: Display alarm only	Reset automatically

Fault code	Fault name	Fault level	Default fault level	Reset method
28-2	Customized alarm 2	0: Blocked 5: Display alarm only	5: Display alarm only	Reset automatically
28-3	Customized alarm 3	0: Blocked 5: Display alarm only	5: Display alarm only	Reset automatically
28-4	Customized alarm 4	0: Blocked 5: Display alarm only	5: Display alarm only	Reset automatically
46-1	The related parameters are set incorrectly.	0: Blocked 5: Display alarm only	0: Blocked	Reset automatically
50-3	FLASH memory error	0: Blocked	0: Blocked	Reset by power-off
59-1	Module 1 temperature too low	0: Blocked	0: Blocked	Reset automatically
59-2	Module 2 temperature too low	0: Blocked	0: Blocked	Reset automatically
63-1	External alarm 1	5: Display alarm only	5: Display alarm only	Reset automatically
63-2	External alarm 2	5: Display alarm only	5: Display alarm only	Reset automatically
63-4	OFF2 activated	0: Blocked 6: No error handling	5: Display alarm only	Reset automatically
82-1	RS485: communication timeout	0: Blocked	0: Blocked	Reset automatically
83-1	Master offline	0: Blocked	0: Blocked	Reset automatically
83-2	Multiple masters existed	0: Blocked	0: Blocked	Reset automatically
83-4	Master/Slave DC bus voltage deviation too large	0: Blocked	0: Blocked	Reset automatically
85-1	Interruption	0: Blocked	0: Blocked	Reset by power-off
117-1	Continuous braking of the brake timeout	0: Blocked	0: Blocked	Reset by power-off

3. Description of Fault Codes

1-1: Current detection circuit error

Default fault level:
0: Blocked

Reset method: Reset manually

Fault level 0: Blocked

range:

Fault cause:

The detected zero drift of the current is too large after the main circuit is switched on.

Troubleshooting:

Cause	Check method	Solution
You can check the current detection value through U5-44 in the stop state.	In the stop state, check whether the current detection value U5-44 is too large.	1. Re-connect the control board to the power module. 2. Replace the power module.
Current detection module error	In the stop state, check whether the current detection value U5-44 is too large.	Replace the power module.
Other causes	-	Contact Inovance for technical support.

2-1: Hardware overcurrent

Default fault 0: Blocked

Reset method: Reset by power-off

level:

Fault level 0: Blocked

range:

Fault cause:

A hardware-triggered overcurrent signal is detected: The instantaneous current upon braking is greater than the hardware overcurrent threshold.

Troubleshooting:

Cause	Check method	Solution
The resistance of the brake is too low.	Check whether the resistance of the braking resistor A5-05 is lower than the minimum recommended value.	Use a braking resistor according to the recommended resistance range.
The braking resistor is short-circuited.	Check whether the braking resistor is short-circuited.	Re-connect the braking resistor correctly.
Other causes	If no interference source exists, the driver board or Hall sensor may be faulty. In this case, replace the faulty driver board or Hall sensor.	Contact Inovance for technical support.

2-2: Software overcurrent

Default fault level: 0: Blocked

Reset method: Reset by power-off/Reset manually

Fault level range: 0: Blocked

Fault cause:

The maximum value of the braking current exceeds the preset software overcurrent threshold defined by A4-09.

Troubleshooting:

Cause	Check method	Solution
A4-09 (Software overcurrent threshold) is set improperly.	Check whether the value of A4-09 is too low.	Set A4-09 properly.
The resistance of the brake is too low.	Check whether the resistance of the braking resistor A5-05 is lower than the minimum recommended value.	Use a braking resistor according to the recommended resistance range.
The braking resistor is short-circuited.	Check whether the braking resistor is short-circuited.	Re-connect the braking resistor correctly.
Other causes	If no interference source exists, the driver board or Hall sensor may be faulty. In this case, replace the faulty driver board or Hall sensor.	Contact Inovance for technical support.

5-1: DC bus voltage too high

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level range: 0: Blocked

Fault cause:

The DC bus voltage exceeds the DC bus overvoltage threshold (A4-11).

Troubleshooting:

Cause	Check method	Solution
The regenerative energy is too high.	Check whether the regenerative energy of the DC bus device is too large.	Select a proper model to ensure that the braking power is met.
The braking resistor is improper.	Check whether the resistance of the braking resistor is between A5-05 and A5-06.	Use a braking resistor according to the recommended resistance range.
The power of the braking resistor is too low.	Calculate whether the system feedback energy is greater than the braking capacity of the brake.	Select a proper model to ensure that the braking power is met.
A5-07 is set improperly.	Check whether the value of A5-07 is too high.	Set A5-07 properly.
The DC bus voltage correction coefficient (A3-13 to A3-16) is improper.	-	Set DC bus voltage correction coefficient (A3-13 to A3-16) properly.
Other causes	-	Contact Inovance for technical support.

10-1: Drive overload

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level range: 0: Blocked

Fault cause:

-

Troubleshooting:

Cause	Check method	Solution
The load is too heavy.	Check whether cumulative overload occurs.	Reduce the load.
The drive power rating is low.	Calculate whether the system feedback energy is greater than the braking capacity of the brake.	Select a proper model to ensure that the braking power is met.
The braking resistor used is improper.	Check whether the resistance of the braking resistor is between A5-05 and A5-06.	Use a braking resistor according to the recommended resistance range.

Other causes	-	Contact Inovance for technical support.
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11-1: Braking resistor overload

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The loading time of A0-02 (RMS value of the braking current) exceeds the preset loading time defined by A6-31...A6-48.

Troubleshooting:

Cause	Check method	Solution
The braking resistor used is improper.	Check whether the resistance of the braking resistor is between A5-05 and A5-06.	1. Configure a braking resistor with the recommended resistance. 2. Contact the agent or Inovance for technical support.

14-1: Module 1 overtemperature

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level 0: Blocked

range:

Fault cause:

IGBT module 1 temperature detection value (U5-08) is higher than the overtemperature threshold (A4-14).

Troubleshooting:

Cause	Check method	Solution
The ambient temperature is high.	Check whether the ambient temperature for installing the brake device is higher than the temperature defined by A4-14.	Lower down the ambient temperature.

The air duct is blocked.	Check whether the fan air duct is clear.	Clean the air duct.
The fan is damaged.	Check whether the fan is damaged.	Replace the fan.
The thermistor is abnormal.	-	Contact Inovance for technical support.
Other causes	-	Contact Inovance for technical support.

14-2: Module 2 overtemperature

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range:

Fault cause:

IGBT module 2 temperature detection value (U5-09) is higher than the overtemperature threshold (A4-14).

Troubleshooting:

Cause	Check method	Solution
The ambient temperature is high.	Check whether the ambient temperature for installing the brake device is too high.	Lower down the ambient temperature.
The air duct is blocked.	Check whether the fan air duct is clear.	Clean the air duct.
The fan is damaged.	Check whether the fan is damaged.	Replace the fan.
The thermistor is abnormal.	-	Contact Inovance for technical support.
Other causes	-	Contact Inovance for technical support.

15-1: External fault 1

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The external fault 1 (15-1) is triggered when H0-00 is set to 1.

Troubleshooting:

Cause	Check method	Solution
The setting source of H0-00 is active.	Check whether H0-00 and the source are active.	Set H0-00 properly.
H0-00 is set improperly.	Check whether H0-00 and the source are active.	1. Set H0-00 to 0. 2. Set H0-00 correctly.

15-2: External fault 2

Default fault 0: Blocked
level:

Reset method: Reset automatically

Fault level 0: Blocked
range: 5: Display alarm only

Fault cause:

External fault 2 (15-2) is triggered when H0-01 is set to 1.

Troubleshooting:

Cause	Check method	Solution
The setting source of H0-01 is active.	Check whether H0-01 and the source are active.	Set H0-01 properly.
H0-01 is set improperly.	Check whether H0-01 and the source are active.	1. Set H0-01 to 0. 2. Set H0-01 correctly.

15-3: Brake enable (DI) not activated

Default fault 0: Blocked
level:

Reset method: Reset
automatically/Reset by
power-off

Fault level 0: Blocked
range:

Fault cause:

When the bus voltage exceeds the undervoltage threshold, the DI1 braking enable signal is inactive.

Troubleshooting:

Cause	Check method	Solution
The DI signal is in inactive state.	Check whether F0-01 is set to 1.	1. Connect the DI terminals properly. 2. Set F0-04 correctly.

22-1: Braking resistance too high

Default fault 0: Blocked

Reset method: Reset by power-off

level:

Fault level 0: Blocked

range:

Fault cause:

When A5-30 is set to 1, the braking resistance is detected to be greater than the maximum value defined by A5-06.

Troubleshooting:

Cause	Check method	Solution
The braking circuit is disconnected.	Check whether the braking resistor is connected properly.	Re-connect the braking resistor.
The braking resistance is too high.	Check whether the resistance of the braking resistor (A5-06) is higher than the maximum recommended value.	Use a braking resistor according to the recommended resistance range.
The braking current sampling is abnormal.	In the stop state, check whether the current detection value U5-44 is too large.	1. Replace the power module. 2. Contact the agent or Inovance for technical support.

22-2 Braking resistance too low

Default fault 0: Blocked

Reset method: Reset by power-off

level:

Fault level 0: Blocked

range:

Fault cause:

Check whether the resistance of the braking resistor is lower than the value of A5-05.

Troubleshooting:

Cause	Check method	Solution
The detected braking resistance is too low.	Check whether the braking resistance is lower than the minimum recommended resistance defined by A5-05.	Use a braking resistor according to the recommended resistance range.
The braking current sampling is abnormal.	In the stop state, check whether the current detection value U5-44 is too large.	1. Replace the power module. 2. Contact the agent or Inovance for technical support.

22-3: Braking resistor detection loss

Default fault 0: Blocked
level:

Reset method: Reset by power-off/Reset
automatically

Fault level 0: Blocked
range:

Fault cause:

When A5-35 is set to 1, the braking current upon pulsing is too low.

Troubleshooting:

Cause	Check method	Solution
The braking circuit is disconnected.	Check whether the braking resistor is connected properly.	Re-connect the braking resistor.
The braking resistance is too high.	Check whether the resistance of the braking resistor (A5-06) is higher than the maximum recommended value.	Use a braking resistor according to the recommended resistance range.
The braking current sampling is abnormal.	In the stop state, check whether the current detection value U5-44 is too large.	1. Replace the power module. 2. Contact the agent or Inovance for technical support.

27-1: Customized fault 1

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The customized fault 1 (27-1) is triggered when H0-08 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-08 is set improperly.	Check whether (H0-08) and the source are active.	1. Set H0-08 to 0. 2. Set H0-08 correctly.
The setting source of H0-08 is active.	Check whether (H0-08) and the source are active.	Set H0-08 properly.

27-2: Customized fault 2

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

Customized fault 2 (27-2) is triggered when H0-09 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-09 is set improperly.	Check whether H0-09 and the source are active.	1. Set H0-09 to 0. 2. Set H0-09 correctly.
The setting source of H0-09 is active.	Check whether H0-09 and the source are active.	Set H0-09 properly.

27-3: Customized fault 3

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

Customized fault 3 (27-3) is triggered when H0-10 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-10 is set improperly.	Check whether H0-10 and the source are active.	1. Set H0-10 to 0. 2. Set H0-10 correctly.
The setting source of H0-10 is active.	Check whether H0-10 and the source are active.	Set H0-10 properly.

27-4: Customized fault 4

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

Customized fault 4 (27-4) is triggered when H0-11 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-11 is set improperly.	Check whether H0-11 and the source are active.	1. Set H0-11 to 0. 2. Set H0-11 correctly.
The setting source of H0-11 is active.	Check whether H0-11 and the source are active.	Set H0-11 properly.

28-1: Customized alarm 1

Default fault 5: Display alarm only

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The customized alarm 1 (28-1) is triggered when H0-13 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-13 is set improperly.	Check whether H0-13 and the source are active.	1. Set H0-13 to 0. 2. Set H0-13 correctly.
The setting source of H0-13 is active.	Check whether H0-13 and the source are active.	Set H0-13 properly.

28-2: Customized alarm 2

Default fault level: 5: Display alarm only

Reset method: Reset automatically

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The customized alarm 2 (28-2) is triggered when H0-14 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-14 is set improperly.	Check whether H0-14 and the source are active.	1. Set H0-14 to 0. 2. Set H0-14 correctly.
The setting source of H0-14 is active.	Check whether H0-14 and the source are active.	Set H0-14 properly.

28-3: Customized alarm 3

Default fault level: 5: Display alarm only

Reset method: Reset automatically

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The customized alarm 3 (28-3) is triggered when H0-15 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-15 is set improperly.	Check whether H0-15 and the source are active.	1. Set H0-15 to 0. 2. Set H0-15 correctly.
The setting source of H0-15 is active.	Check whether H0-15 and the source are active.	Set H0-15 properly.

28-4: Customized alarm 4

Default fault 5: Display alarm only

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

The customized alarm 4 (28-4) is triggered when H0-16 is set to 1.

Troubleshooting:

Cause	Check method	Solution
H0-16 is set improperly.	Check whether H0-16 and the source are active.	1. Set H0-16 to 0. 2. Set H0-16 correctly.
The setting source of H0-16 is active.	Check whether H0-16 and the source are active.	Set H0-16 properly.

46-1: Parameter setting incorrect

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range: 5: Display alarm only

Fault cause:

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Troubleshooting:

Cause	Check method	Solution
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Fault information 1: The current at each point of the braking resistor overload curve is not incremental.	Check whether the resistor overload curve point-current effective values given in A06-31, A06-34, A06-37, A06-40, A06-43, and A06-46 are incremental.	Set A06-31, A06-34, A06-37, A06-40, A06-43, and A06-46 properly.
Fault information 2: The maximum braking rate of channel 2 braking curve (A6.06) is lower than the initial braking rate (A6.02).	Check whether the value of A6-06 is higher than or equal to A6-02.	Set A6-06 and A6-02 properly.
Fault information 3: The maximum braking rate voltage threshold of channel 2 braking curve (A6.04/A6.05) is lower than the starting voltage (A6.01).	Check whether the value of A6-05 is higher than or equal to A6-01.	Set A6-05 and A6-01 properly.

50-3: FLASH memory error

Default fault 0: Blocked
level:

Reset method: Reset by power-off

Fault level 0: Blocked
range:

Fault cause:

The internal FLASH memory cannot be written or erased properly.

Troubleshooting:

Cause	Check method	Solution
FLASH memory is abnormal.	-	Contact Inovance for technical support.
Other causes	-	Contact Inovance for technical support.

59-1: Module 1 temperature too low

Default fault 0: Blocked
level:

Reset method: Reset automatically

Fault level 0: Blocked

range:

Fault cause:

IGBT module 1 temperature detection value (U5-08) is lower than the low temperature alarm threshold (A4-16).

Troubleshooting:

Cause	Check method	Solution
The ambient temperature is too low.	Check whether the ambient temperature for installing the brake device is lower than the temperature defined by A4-16.	Increase the ambient temperature.
The low temperature alarm threshold (A4-16) is too high.	Check whether the value of A4-16 is too high.	Set the low temperature detection threshold (A4-16) properly.
The thermistor is abnormal or the thermistor circuit is disconnected.	-	Contact Inovance for technical support.
The module is damaged.	-	Contact Inovance for technical support.

59-2: Module 2 temperature too low

Default fault 0: Blocked

Reset method: Reset automatically

level:

Fault level 0: Blocked

range:

Fault cause:

IGBT module 2 temperature detection value (U5-09) is lower than the low temperature alarm threshold (A4-16).

Troubleshooting:

Cause	Check method	Solution
The ambient temperature is too low.	Check whether the ambient temperature for installing the brake device is lower than the temperature defined by A4-16.	Increase the ambient temperature.
The low temperature alarm	Check whether the value of A4-16 is too	Set the low temperature detection

threshold (A4-16) is too high.	high.	threshold (A4-16) properly.
The thermistor is abnormal or the thermistor circuit is disconnected.	-	Contact Inovance for technical support.
The module is damaged.	-	Contact Inovance for technical support.

63-1: External alarm 1

Default fault level: 5: Display alarm only

Reset method: Reset automatically

Fault level range: 5: Display alarm only

Fault cause:

The external alarm 1 (63-1) is triggered when H0-05 is set to 1.

Troubleshooting:

Cause	Check method	Solution
The setting source of H0-05 is active.	Check whether H0-05 and the source are active.	Set H0-05 properly.
H0-05 is set improperly.	Check whether H0-05 and the source are active.	1. Set H0-05 to 0. 2. Set H0-05 correctly.

63-2: External alarm 2

Default fault level: 5: Display alarm only

Reset method: Reset automatically

Fault level range: 5: Display alarm only

Fault cause:

The external alarm 2 (63-2) is triggered when H0-06 is set to 1.

Troubleshooting:

Cause	Check method	Solution
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H0-06 is set improperly.	Check whether H0-06 and the source are active.	Set H0-06 properly.
The setting source of H0-06 is active.	Check whether H0-06 and the source are active.	1. Set H0-06 to 0. 2. Set H0-06 correctly.

63-4: OFF2 activated

Default fault level: 5: Display alarm only

Reset method: Reset automatically

Fault level 0: Blocked

range: 6: No error handling

Fault cause:

If the OFF2 signal set by A5-01 or A5-02 is activated (OFF2 = 0), the OFF2 activation alarm 63-4 is triggered.

Troubleshooting:

Cause	Check method	Solution
The OFF2 signal set by A5-01 or A5-02 has been activated.	Check whether the OFF2 signal set by A5-01 or A5-02 cannot be braked.	Set A5-01 and A5-02 properly.

82-1: RS485 communication timeout

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level 0: Blocked

range:

Fault cause:

RS485 communication has no received data or received data CRC check error for a continuous period exceeding the preset timeout time set by A10-06.

Troubleshooting:

Cause	Check method	Solution
Timeout occurred during normal communication between the braking unit and external device.	Check whether the communication link between the braking unit and the external device is connected properly.	1. Re-connect the communication link. 2. Contact the agent or Inovance for technical support.

83-1: Master offline

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level range: 0: Blocked

Fault cause:

When the device is a slave, the master/slave connection is not established.

Troubleshooting:

Cause	Check method	Solution
The slave fails to receive the synchronization signals from the master.	1. Check whether the main and slave CAN networks are connected properly. 2. Check whether the master and slave baud rates (A5-40) are consistent. 3. Check whether CANH and CANL are connected reversely. 4. Check whether multiple masters exist in the network. 5. Check whether the master and slave CAN network termination resistances are connected properly.	1. Set A5-40 properly. 2. Connect the CAN communication circuit properly. 3. Set a unique address for the master and slave. 4. Set DIP switches 1 and 2 of S2 on the first and the last units to ON when more than two braking units are connected in parallel. 5. Contact the agent or Inovance for technical support.

83-2: Multiple masters

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level range: 0: Blocked

Fault cause:

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Troubleshooting:

Cause	Check method	Solution
Multiple master exist in the	Check whether multiple masters exist	Set A5-43, A5-44, and S4 DIP switch

network.	in the master/slave (A5-43 and A5-44), or S4 DIP switch setting.	properly.
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83-4: Master/Slave DC bus voltage deviation too large

Default fault level: 0: Blocked

Reset method: Reset automatically

Fault level range: 0: Blocked

Fault cause:

The DC bus voltage deviation of the master and slave exceeds the threshold defined by A05-49.

Troubleshooting:

Cause	Check method	Solution
The slave detects that the deviation between the slave DC bus voltage and master DC bus voltage is too large.	1. Check that the master and slave are in the same DC bus network. 2. Check whether the DC bus voltage difference of the master and slave devices under the same bus is greater than A5-49.	1. Re-connect the master and slave braking topology. 2. Perform DC bus voltage correction again on A3-13 to A3-16. 3. Contact the agent or Inovance for technical support.

85-1: ADC interruption error

Default fault level: 0: Blocked

Reset method: Reset by power-off

Fault level range: 0: Blocked

Fault cause:

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Troubleshooting:

Cause	Check method	Solution
The software internal execution is abnormal.	-	Power off and on again.

Other causes	-	Contact Inovance for technical support.
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117-1: Continuous braking of the brake timeout

Default fault level:
0: Blocked

Reset method: Reset by power-off

Fault level range:
0: Blocked

Fault cause:

When the allowable continuous braking time (A5-29) is not 0, the actual braking duration exceeds this value, triggering the continuous braking timeout fault.

Troubleshooting:

Cause	Check method	Solution
A5-29 (Continuous braking time) is set improperly.	Check whether A5-29 is set to an excessively low non-zero value.	Set A5-29 properly.
The brake ON threshold (A5-07) is set to an excessively low value.	Check whether the value of A5-07 is too low.	Set A5-07 properly.
The brake stop hysteresis voltage (A5-08) is set to an excessively high value.	Check whether the value of A5-08 is too high.	Set A5-08 properly.
The power of the brake selected is too low and the load energy is too high.	Calculate whether the system feedback energy is greater than the braking capacity of the brake.	Select a proper model to ensure that the braking power is met.
The DC bus voltage has not be calibrated, leading to large difference from the actual bus voltage.	Check that A3-13 to A3-16 are set properly.	Perform DC bus voltage correction again on A3-13 to A3-16.
Other causes	-	Contact Inovance for technical support.



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